



VPL: A Novel Cost Model for Earth and Mars-Orbit Navigation and Communication Constellations

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The problem

Limited public unit-cost data for navigation and communication constellations. Legacy models (SMAD USCM8, QuickCost) rest on data up to the early 2000s. We build a cost model on **27 systems** (1997–2025), that also covers New-Space systems.

The resolved CER model

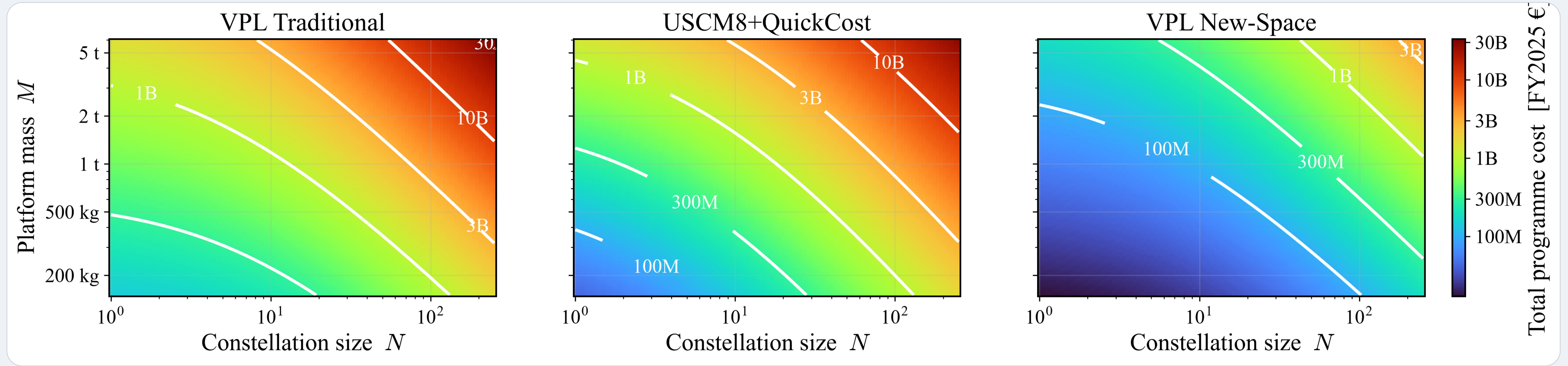
$$COST(M, N) = F \cdot g \cdot \left[\underbrace{DEV(M)}_{0.215 \cdot M^{0.828}} + \underbrace{PFM(M) \cdot N^{LRE}}_{3.29 \cdot M^{0.682}} \right]$$

- M = platform mass [kg],
- N = units, cost in FY2025 €M.
- $LRE = 1 + \ln(0.9)/\ln(2) = 0.848$
- $g = 1$ traditional, 0.118 New-Space.
- $F = 1$ Earth, 1.07 Mars.

Main metrics

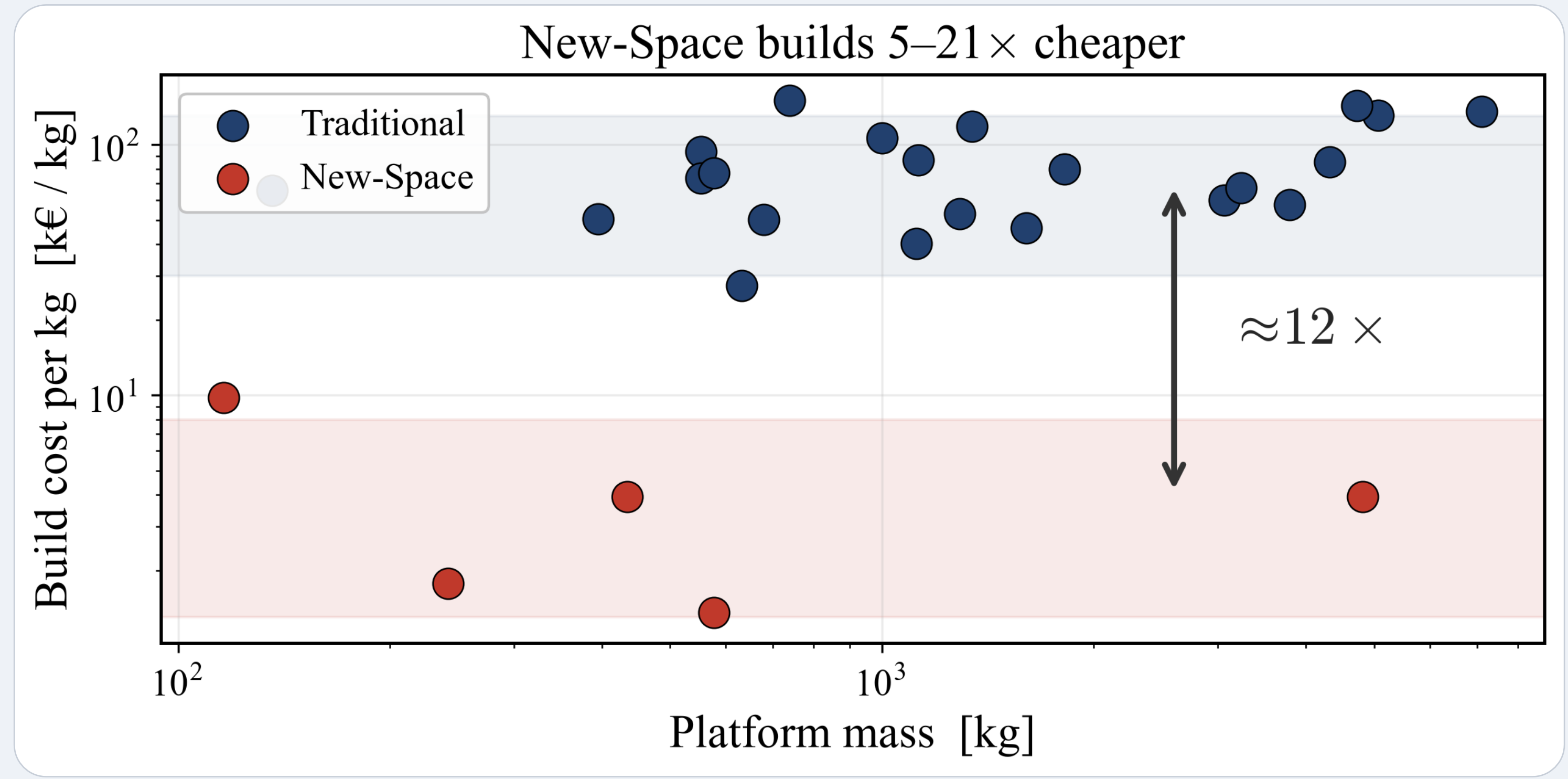
0.92 cross-validated R^2
5–21× New-Space cheaper
 $\approx$ USCM8+QuickCost on traditional;
but SMAD over-predicts New-Space $\sim 8\times$

Cost vs mass and constellation size



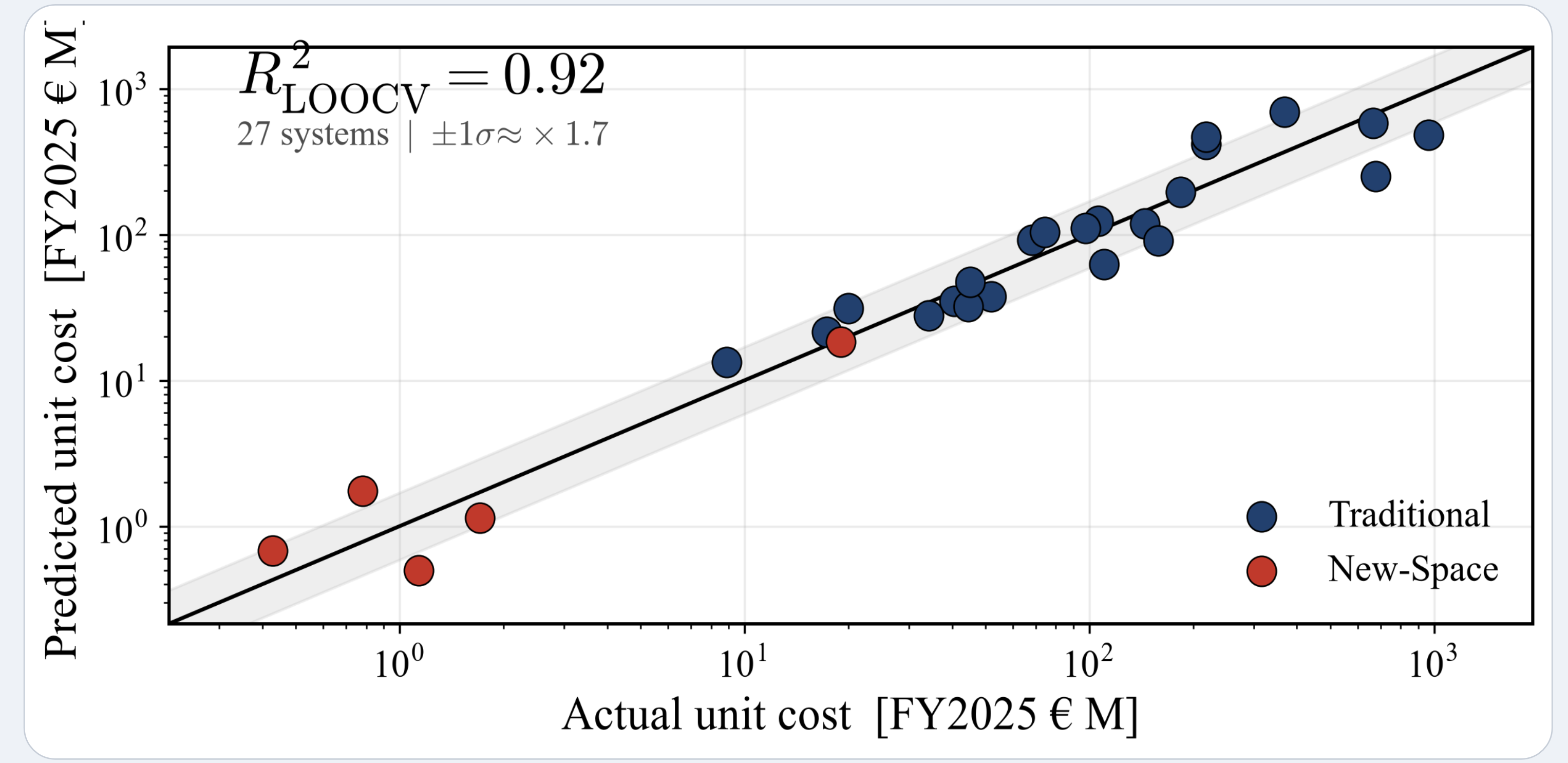
Iso-cost contours of total program cost $TOTAL(M,N)$. VPL Traditional tracks the SMAD benchmark closely, with VPL New-Space a full cost band below.

Cost-Plus vs New-Space Economy



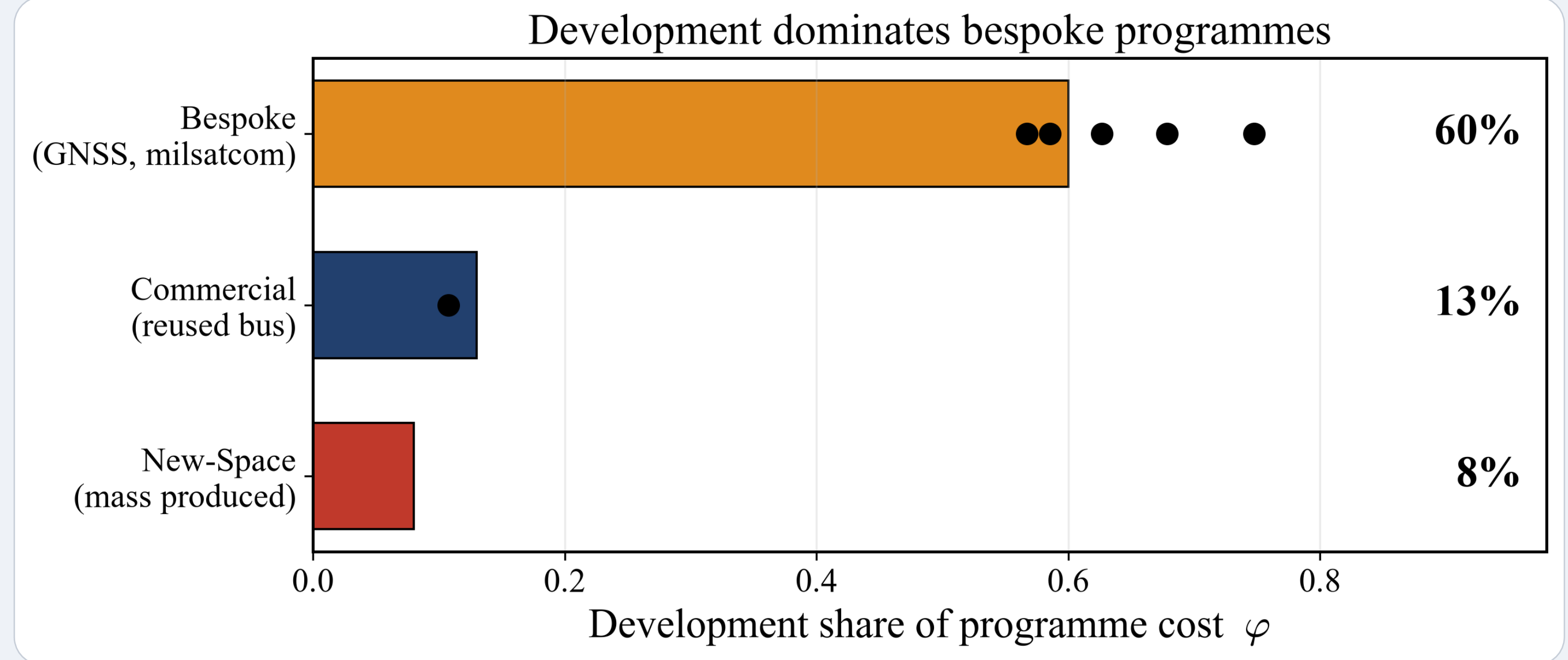
Mass-produced systems are 5–21× cheaper than traditional and bespoke ones.

Validation



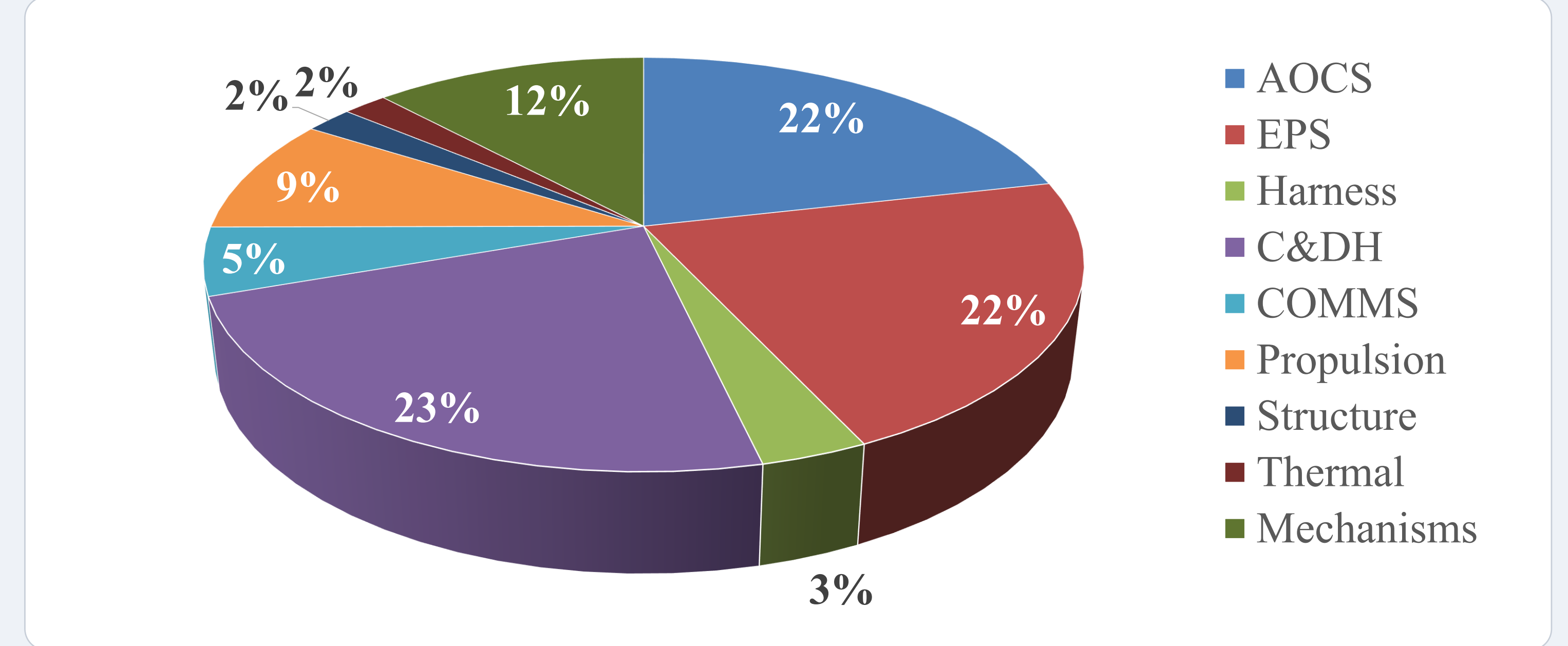
CER tracks cost across 3 orders of magnitude ($R^2 = 0.81$ for traditional-only).

Development vs production



The DEV/PFM split fixed by disclosed programme budgets (US SARs, ESA, GAO).

Planetary cost factor



Total planetary factor : $7.2\% \pm 4.5\%$ (Mars bus-cost increase)

Data and code



GitHub repository with the dataset, per-number sources, the Python model, the reproduction notebook, this poster and a detailed report.
github.com/cholevasm/vpl-cost-model

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